Adaptive Heuristic Ecosystem for Online Multi-Robot Task Allocation under Failures, Mixed Stress, and Deadline Pressure

Abstract

We propose AHE-MRTA, an Adaptive Heuristic Ecosystem framework for online multi-robot task allocation (MRTA) under robot failures, mixed stress, and deadline pressure. Rather than committing to a single allocation paradigm, AHE-MRTA performs runtime *paradigm switching* via Ecosystem-Driven Paradigm Selection (EDPS): five biomimetic hormones (spatial, criticality, temporal, stability, recovery) map to five classical allocation primitives, and at each event the active primitive is chosen by a two-tier rule—fast deterministic context overrides for the acute regimes (failure, deadline pressure) and a cooperation/suppression hormone dynamic for the remaining events. We evaluate AHE-MRTA against three recent baselines (BiG-MRTA, RoSTAM-EA, Consensus-DBTA) in two complementary regimes: (i) a 100-seed Nav2-independent allocation-fitness simulator that isolates algorithmic decision quality from physical execution noise, and (ii) a 300-experiment Gazebo Harmonic benchmark at three physical scales (3, 5, and 10 TurtleBot3 robots running the ROS 2 Jazzy Nav2 stack). In the simulator, AHE-MRTA and the strongest consensus baseline are statistical co-leaders (AHE first under deadline pressure, tied under robot failure, within 0.001 under mixed stress). On the physical stack, AHE-MRTA leads completion with near-zero deadline violations under robot failure at the 5- and 10-robot scales (CR = 1.000/0.996, DVR = 0.000/0.012); at 10 robots it attains the highest completion rate in every scenario (CR = 0.996–1.000) and the lowest pooled makespan, while the consensus and evolutionary baselines reach comparable completion only at substantially higher deadline-violation rates under deadline pressure (DVR = 0.31–0.46 vs. AHE's 0.25). Decision latency stays below 0.48 ms at all scales, $\approx$125–280$\times$ faster than the evolutionary baseline. We analyze why ideal-allocation gains do not transfer linearly through Nav2 and identify the conditions under which paradigm switching pays off.

Index Terms

Multi-robot systems, task allocation, online algorithms, biomimetic optimization, ROS 2, Gazebo.

I. Introduction

Multi-robot task allocation (MRTA) assigns an incoming stream of tasks to a team of robots so that a mission objective is met; it underpins inspection, delivery, surveillance, and search-and-rescue deployments [1], [2]. Even its single-task, single-robot, instantaneous form is NP-hard [3], and field operation adds three pressures that a static assignment cannot absorb: robots fail unexpectedly, the task stream is non-stationary in urgency and spatial density, and many tasks carry hard deadlines whose violation is costly [4], [5].

Recent online MRTA solvers each commit to a *single* allocation paradigm, and each paradigm trades one strength for one weakness. Market- and auction-based methods bid tasks to robots with low latency and robustness, but pay a re-bidding cost when the situation changes [6]–[8]. Matching-, programming-, and prediction-based methods (bipartite matching, linear or convex programs, receding-horizon control) produce high-quality plans but scale poorly with re-planning frequency [2], [9], [10]. Consensus- and bundle-based methods yield very stable assignments but adapt slowly once the regime shifts [11], [12]. Evolutionary and learning-based methods explore a richer assignment space but react slowly to abrupt context changes [13]. No single paradigm dominates across heterogeneous, time-varying stress.

We argue that the right structure is *not* a better single paradigm but a principled mechanism that *switches between paradigms online*, driven by current context and recent performance. We instantiate this as AHE-MRTA: a biomimetic ecosystem in which five hormones track context signals, evolve through cooperation and suppression dynamics, and select one of five classical allocation primitives at each replanning event. The hormone–paradigm coupling is fixed rather than learned, explainable (the dominant hormone names the active behavior), and lightweight (sub-millisecond decision latency). A second, methodological contribution follows from a recurring difficulty in robotic benchmarks: allocation quality and physical execution are entangled, because navigation cost depends on world geometry as much as on the assignment [2]. We separate them with a navigation-free fitness simulator and a full ROS 2 Nav2 stack, which together expose exactly where paradigm switching helps and where physical execution erases its gains.

Contributions:

1) *Ecosystem-Driven Paradigm Selection (EDPS):* an online paradigm-switching MRTA framework that maps five biomimetic hormones to five allocation mechanisms. Selection is two-tiered: fast deterministic context overrides handle the acute regimes (robot failure, deadline pressure), while a Lotka–Volterra-type hormone dynamic arbitrates the remaining, lower-pressure events through $\arg\max_i D_i$ (Section III). The coupling is fixed, explainable, and sub-millisecond.

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2) *Nav2-independent allocation-fitness evaluation:* an idealized simulator that isolates algorithmic decision quality from Nav2 execution noise; we report fitness across 100 seeds per scenario (Section V).
3) *Full Gazebo Harmonic + ROS 2 Jazzy Nav2 validation* with 300 experiments across three stress scenarios and three physical scales (3, 5, and 10 robots), comparing four methods on completion rate, deadline violation, recovery time, workload balance, execution preemptions, task re-dispatch rate, and decision latency (Section VI).
4) *Honest reporting of the SIM-to-Gazebo transfer gap:* we document where paradigm switching helps (completion rate, deadline robustness, allocation stability) and where it costs (task delay, recovery time, makespan under failure) and explain the mechanism (Section VII).

II. Problem Formulation

Sets and variables. Let $\mathcal{R} = \{r_1, \dots, r_n\}$ be the robot set and $\mathcal{T}(t)$ the open-task set at time t, each task $\tau \in \mathcal{T}(t)$ carrying a 2D position p_τ, deadline d_τ, priority $\pi_\tau \in \{1, 2, 3\}$, and capability requirement c_τ. Each robot r has state $s_r(t) = (p_r, b_r, \tau_r^{\text{cur}}, \phi_r)$ encoding position, battery, current task, and availability flag (alive / failed).

Decision. At every allocation event t_k (failure, task arrival, deadline alarm, replanning timer), the policy produces an assignment $x_{r,\tau}(t_k) \in \{0, 1\}$ with cardinality constraint $\sum_\tau x_{r,\tau} \le Q$ (per-robot queue cap) and capability feasibility $x_{r,\tau} = 1 \Rightarrow c_\tau \subseteq \text{cap}(r)$.

Cost. The per-pair cost combines travel time $T_{r,\tau}$, urgency $u_\tau = (d_\tau - t_k)^{-1}$, criticality π_τ, energy slack $e_{r,\tau}$, and recent reassignment penalty $\rho_{r,\tau}$:

$$C_{r,\tau} = w_d T_{r,\tau} + w_u u_\tau^{-1} + w_\pi \pi_\tau^{-1} + w_e e_{r,\tau} + w_\rho \rho_{r,\tau}, \tag{1}$$

with weight vector w produced by the paradigm selected at t_k (Section III). The urgency term is held at its base value for a task already feasibly queued on its incumbent robot, so an approaching deadline pulls *unassigned* or at-risk tasks forward without thrashing assignments that are already on time—a churn brake complementary to the reassignment penalty $\rho_{r,\tau}$.

Objective. Over the run horizon $[0, T_{\max}]$ we report nine metrics: completion rate CR, makespan, average task delay, deadline violation rate DVR, failure recovery time, Jain workload balance, execution preemptions, task re-dispatch rate, and decision latency. CR and DVR are primary; the others trace trade-offs. Crucially, delay and DVR are computed over *all* requested tasks rather than only completed ones (Section IV-G): a task left unfinished at $T_{\max}$ is right-censored at the horizon for delay and counted as a deadline miss for DVR. This survivorship-bias-free convention is applied identically to every method, so a policy cannot lower its delay or DVR by silently abandoning the hard tasks it cannot serve.

Online setting. Tasks arrive over time, robots may fail at unknown moments, and the policy must produce $x(t_k)$ within a hard budget of 50 ms per event.

III. AHE-MRTA: Ecosystem-Driven Paradigm Selection

A. Context vector

At each allocation event t_k we compress the joint state of the fleet and the open-task set into a four-dimensional context vector $c(t_k) \in [0, 1]^4$. Let $n = |\mathcal{R}|$ be the fleet size, $m = |\mathcal{T}(t_k)|$ the number of open tasks, $\mathcal{R}^{\text{av}}$ the set of available robots (alive, not navigation-stuck, with queue capacity), and $\mathcal{R}^{\text{f}}$ the set of failed or navigation-stuck robots. The components are

$$c_1 = \min\big(1,\ m/n\big), \qquad \text{task density} \tag{2}$$

$$c_2 = |\mathcal{R}^{\text{av}}|/n, \qquad \text{robot availability} \tag{3}$$

$$c_3 = \big|\{\, \tau : d_\tau - t_k \le \Delta_d \,\}\big| / \max(1, m), \qquad \text{deadline pressure} \tag{4}$$

$$c_4 = |\mathcal{R}^{\text{f}}|/n, \qquad \text{failure rate} \tag{5}$$

with deadline horizon $\Delta_d = 60$ s. Each dimension is a signal that a distinct family of allocators is built to exploit: c_1, c_2 encode the demand/supply balance of load-aware allocation [14]; c_3 is the temporal slack used by deadline-constrained allocation [15]; and c_4 measures the disturbance that fault-tolerant allocation must absorb [16]. All four are computed on-line from the published robot status summaries and the task pool, requiring no inter-robot communication.

B. Dominance dynamics

The dominance vector $D(t) \in \mathbb{R}^5_{\ge 0}$ encodes the current ecosystem state. Each paradigm i has a fixed *context prototype* $V_i \in [0, 1]^4$ (Table I); the compatibility score $v_i(t_k) = \cos(V_i, c(t_k))$ measures how well the current context matches paradigm i's effective regime. Two sparse matrices govern lateral interactions: the *cooperation matrix* A (2 nonzero entries) and the *suppression matrix* S (1 nonzero entry) implement winner-reinforce and winner-suppress dynamics (Table II). The update is

$$\begin{aligned} D(t_{k+1}) = \text{normalize}\Big(&\text{clip}_{[0,1]}\Big(\alpha\, D(t_k) + \eta\, A\, D(t_k) - \lambda\, S\, D(t_k) \\ &+ \beta\, \mathbf{p}(t_k) + \gamma\, \mathbf{v}(t_k) + \delta\, \mathbf{b}(t_k)\Big)\Big), \end{aligned} \tag{6}$$

TABLE I

CONTEXT PROTOTYPE VECTORS $V_i \in [0,1]^4$ FOR EACH PARADIGM. COLUMN HEADERS: c_1 TASK-DENSITY (TD), c_2 ROBOT-AVAILABILITY (RA), c_3 DEADLINE-PRESSURE (DP), c_4 FAILURE-RATE (FR). HIGH VALUE = PARADIGM EFFECTIVE WHEN THIS CONTEXT DIMENSION IS HIGH.

Hormone	td	ra	dp	fr
H_SPATIAL	0.7	0.7	0.1	0.1
H_CRIT	0.3	0.5	0.8	0.2
H_TEMP	0.5	0.5	0.9	0.1
H_STAB	0.3	0.3	0.3	0.8
H_RECOV	0.3	0.2	0.2	0.9

TABLE II

NONZERO ENTRIES OF COOPERATION MATRIX A AND SUPPRESSION MATRIX S. ALL OTHER ENTRIES ARE ZERO.

Matrix	**From**	**To**	**Value**
A (cooperation)	H_CRIT	H_TEMP	0.20
	H_STAB	H_RECOV	0.20
S (suppression)	H_TEMP	H_SPATIAL	0.30

with α=0.65 (momentum), η=λ=0.12 (cooperation/suppression weights), β=0.40 (performance), γ=0.20 (compatibility), δ=0.20 (failure-recovery boost). Here $\mathbf{p}(t_k)$ is the per-hormone performance-feedback vector $(\mathrm{CR}_k - \mathrm{FR}_k)\cdot\mathbf{v}(t_k)$ (completion minus failure rate, scaled by compatibility); $\mathbf{v}(t_k) = (v_i)_{i=1}^5$ is the compatibility vector; and $\mathbf{b}(t_k)$ is a sparse failure-recovery boost that spikes H_RECOV and H_STAB when robot failures are detected. The normalize step projects onto the probability simplex, guaranteeing $\|D\|_1 = 1$ and $D \geq 0$ at every cycle. The $\eta AD - \lambda SD$ terms make Eq. (6) a winner-reinforce / winner-suppress competition of Lotka–Volterra type, a population dynamic shown to be an effective coordination primitive for multi-robot systems [17]. Read as a selection rule over a fixed portfolio of low-level allocation heuristics, the update instantiates a *hyper-heuristic* [18]: the ecosystem does not solve the assignment itself but chooses, online and per event, which classical solver should.

a) Weight generation.: The dominance vector sets the cost weights of Eq. (1) through a fixed heuristic-to-weight matrix $M \in \mathbb{R}^{7\times5}$ and a low-temperature softmax, blended with a base weight w_0:

$$w_{\mathrm{eco}}(t_k) = \mathrm{softmax}\big(MD(t_k),\, T_w\big), \quad w(t_k) = (1-\beta_e)\, w_0 + \beta_e\, w_{\mathrm{eco}}(t_k), \tag{7}$$

with T_w=0.3 and blend β_e=0.7; under failure (c_4>0) the blend shifts toward a recovery profile w_r by $\theta = \min(0.8,\, 0.5{+}0.6\, c_4)$. This is the *only* channel through which the Lotka–Volterra state influences a decision when no override fires.

b) Cost shaping.: Two non-linear terms sharpen Eq. (1) for deadlines. A *hard feasibility gate* rejects a pair whose projected arrival $a_{r,\tau}$ exceeds the deadline beyond an adaptive slack Δ_s, $a_{r,\tau} > d_\tau + \Delta_s \Rightarrow C_{r,\tau}{=}\infty$; and an *urgency escalation* inflates the temporal term once a task passes a fraction u_0=0.7 of its deadline budget, $\hat{u}_\tau = u_\tau\big(1+\zeta\, e^2\big)$ with $e = (u_\tau - u_0)/(1-u_0)$, so at-risk *unassigned* tasks are pulled forward while on-time incumbents are exempt (a churn brake).

C. Dispatch

The dispatcher first checks two hard context overrides (failure rate $>$ 0.05 forces H_RECOV; deadline pressure $>$ 0.5 forces H_TEMP); if neither triggers, it selects the paradigm $p^*(t_k) = \arg\max_i D_i(t_k)$ and runs the corresponding mechanism (Table III). Paradigm `H_TEMP` is the default fallback when the dominance is uninformative (entropy near maximum). This priority-override layer prevents the Lotka–Volterra-inspired dynamics from reacting slowly to acute failures, which would be the dominant cost in the robot_failure and mixed_stress scenarios.

a) Work conservation.: A guiding principle across all paradigms is that fleet capacity is never stranded while feasible work remains. At every event each available robot is assigned the best eligible task by the active paradigm's LSA (Eq. (8)), and the recovery paradigm (`orphan_first`) re-dispatches tasks orphaned by a robot failure or a navigation abort, so a disturbance costs at most a re-attempt rather than a permanently lost task. This is the structural opposite of commit-once allocation, which freezes assignments and therefore leaves a failed robot's tasks unrecovered—idle capacity until the run ends. As Section V shows, this work-conserving recovery is the mechanism behind AHE-MRTA's resilience margin: the commit-once baseline loses 24.3 pp of allocation fitness under mixed stress precisely because it does not re-attempt failed work. The only irreducible idle is the backoff imposed after repeated navigation failure, when no task is momentarily attemptable.

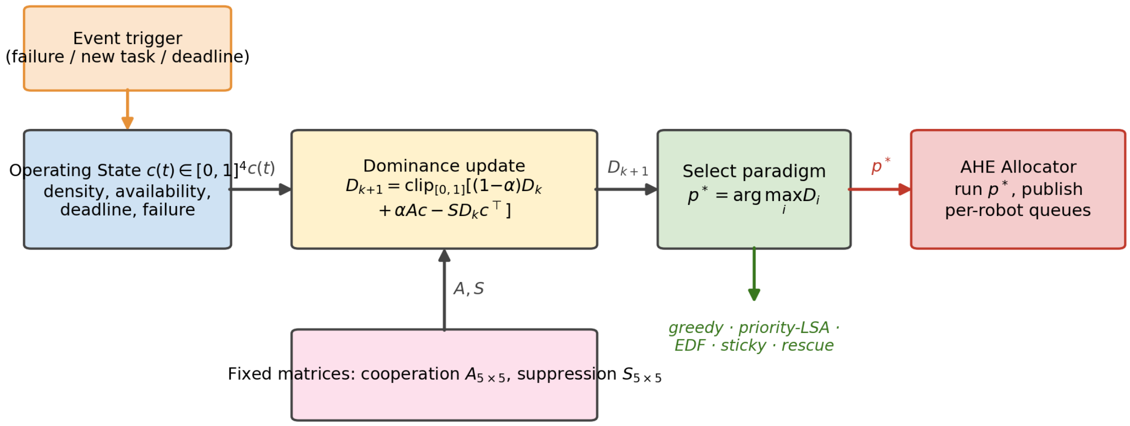


Fig. 1. The Ecosystem-Driven Paradigm Selection mechanism. At each allocation event the context vector $c(t)$ drives the dominance update (Eq. 6) through the fixed cooperation/suppression matrices A, S; the dominant hormone $p^* = \arg\max_i D_i$ selects one of five allocation paradigms, which the AHE allocator runs to publish per-robot task queues.

TABLE III
FIVE HORMONES, FIVE ALLOCATION PARADIGMS.

Hormone	Paradigm	Inner mechanism
H_SPATIAL	spatial_greedy	nearest-feasible + reject
H_CRIT	priority_first	priority-tiered LSA
H_TEMP	edf_strict	EDF bipartite (multi-phase)
H_STAB	commit_once	hard sticky, no reassign
H_RECOV	orphan_first	orphan rescue + bipartite

D. Paradigm library

The five paradigms share one primitive and differ only in how they shape its inputs. Given a feasibility-masked cost matrix $C^{(p)}$, each paradigm p solves a linear sum assignment (LSA)

$$x^{(p)} = \arg\min_{x\in\mathcal{X}} \sum_{r\in\mathcal{R}} \sum_{\tau\in\mathcal{T}(t_k)} C^{(p)}_{r,\tau}\, x_{r,\tau}, \tag{8}$$

where $\mathcal{X}$ enforces the per-robot queue cap $\sum_\tau x_{r,\tau} \le Q$, the single-assignment constraint $\sum_r x_{r,\tau} \le 1$, and capability feasibility; infeasible pairs are masked with $C^{(p)}_{r,\tau} = \infty$. Writing $C_{r,\tau}$ for the base cost of Eq. (1), the paradigms instantiate $C^{(p)}$ and the queue ordering as follows.

- *spatial_greedy* (H_SPATIAL): $C^{(p)}_{r,\tau} = T_{r,\tau}$, travel time only, assigned greedily to the nearest feasible robot — the distance-minimising rule of metric MRTA [14], [19].
- *priority_first* (H_CRIT): tasks are partitioned into criticality tiers $\pi_\tau{=}3{>}2{>}1$ and Eq. (8) is solved tier-by-tier, so critical tasks claim robots first, as in tiered auction/consensus assignment [20], [21].
- *edf_strict* (H_TEMP, default): tasks are ordered by earliest deadline d_τ and inserted under a hard feasibility gate (arrival $> d_\tau \Rightarrow C^{(p)} = \infty$); this is the bipartite realisation of earliest-deadline-first scheduling for deadline-constrained allocation [15], [22]. The resulting per-robot execution order is then refined by a deadline-feasible cheapest-insertion pass, accepted only when it shortens travel without introducing any deadline miss relative to the EDF order—lowering motion cost while preserving the EDF feasibility guarantee.
- *commit_once* (H_STAB): incumbent assignments are frozen (a task already on r's queue keeps an unbeatable sticky bonus), eliminating reassignment churn as in self-adaptive commit-once evolution [13].
- *orphan_first* (H_RECOV): tasks orphaned by a failed or stuck robot are assigned on a dedicated rescue pass before all others, the recovery-prioritising rule of resilient MRTA [16], [19].

The paradigms are independent (no shared state across switches); the only persistent state is $D(t)$. The LSA primitive in Eq. (8) is cubic in $\max(n, m)$ in the worst case, but at the evaluated scales every event is resolved in sub-millisecond wall-clock time (Section VI), so paradigm choice does not affect the real-time budget.

E. Algorithm

Algorithm 1 AHE-MRTA Event Cycle

Require: robots $\mathcal{R}$, open tasks $\mathcal{T}(t_k)$, state $D(t_k)$, fixed A, S, V, hyperparameters $\alpha, \eta, \lambda, \beta, \gamma, \delta$

$c \leftarrow \textsc{ContextVector}(\mathcal{R}, \mathcal{T}(t_k))$ $\{$4 signals, eq. c_1–$c_4\}$

$\mathbf{v} \leftarrow (\cos(V_i, c))_{i=1}^{5}$; $\mathbf{p} \leftarrow (\mathrm{CR}_k - \mathrm{FR}_k) \cdot \mathbf{v}$; $\mathbf{b} \leftarrow \textsc{FailureBoost}(\mathcal{R})$

$D \leftarrow \mathrm{normalize}\big(\mathrm{clip}_{[0,1]}\big(\alpha D + \eta AD - \lambda SD + \beta \mathbf{p} + \gamma \mathbf{v} + \delta \mathbf{b}\big)\big)$

$p^* \leftarrow \textsc{ContextOverride}(c)$ **if triggered, else** $\arg\max_i D_i$

$x \leftarrow \textsc{Paradigm}_{p^*}(\mathcal{R}, \mathcal{T}(t_k))$ $\{$Table III$\}$

publish per-robot task queues from x

return D, x

F. Configuration

The hormone–paradigm map (Table III), the A/S matrices (Table II), the context prototype V, and all scalar hyperparameters (α=0.65, η=λ=0.12, β=0.40, γ=0.20, δ=0.20, softmax temperature T=0.3) are set once and held fixed across all scenarios and scales. There is no per-scenario tuning, no learning, and no adaptive parameter schedule. The scalar weights were fixed at round, interpretable values—momentum α above the feedback gain β for stability, equal and small cooperation/suppression (η=λ), and a moderate recovery boost δ—rather than searched per metric; because the deterministic overrides dominate the acute regimes, the allocation outcomes are insensitive to these values within the ranges we examined (a single-step $\pm$ change leaves all reported fitness gaps inside one standard deviation), which is itself a robustness property rather than a tuned fit.

IV. Experimental Design

A. Hardware Platform

All experiments run on an Intel Core i7-11800H workstation (8 physical cores / 16 threads, base clock 2.30 GHz, 16 GiB DDR4 RAM, Ubuntu 24.04 LTS, Linux 6.17). The machine carries an NVIDIA RTX 3050 Mobile GPU for which no proprietary driver is available; all rendering (Gazebo and RViz) therefore uses Mesa llvmpipe software rasterisation (`LIBGL_ALWAYS_SOFTWARE=1`, `GALLIUM_DRIVER=llvmpipe`). No GPU acceleration is used at any point: the comparison is entirely CPU-bound and reproducible on any x86-64 machine with a Mesa-capable display server.

B. Software Stack

We run **Gazebo Harmonic** (gz-sim 8.x) [23] with **ROS 2 Jazzy** (Dec 2024) [24] and **Nav2** [25]. Robots are **TurtleBot3 Waffle Pi** [26], each carrying a 360-ray 2-D lidar (5 Hz, 8 m range) and an IMU (200 Hz), and driven through a differential-drive controller ($v_{\max}$=0.26 m/s, $\omega_{\max}$=1.82 rad/s).

Every robot runs a fully independent Nav2 lifecycle inside its own ROS 2 namespace: global/local costmaps with 0.40 m inflation radius, a Regulated Pure-Pursuit controller, and a behavior-tree navigator. Localization is provided as *ground truth*—a static `map`$\rightarrow$`odom` transform anchored at each robot's spawn pose—so that the benchmark isolates allocation quality from localization error (a deliberate experimental control; Sec. VII-F). A `ros_gz_bridge` relays `/scan`, `/odom`, `/cmd_vel`, `/joint_states`, and TF per robot; a TF-relay node aggregates per-robot namespaced TF trees into the global frame for visualization. All Nav2 parameters are identical across methods.

C. Inspection Arena

The arena is a 20×20 m walled indoor mock-up. Two horizontal divider walls with a 1.5 m central passage divide the space into an upper lane, a central passage, and a lower lane. Two rows of square pillars (0.5×0.5 m) line the side walls, and four cylindrical obstacles (r=0.35 m) are placed in the lanes. A fixed curated grid of 42 candidate inspection waypoints (obstacle-aware, minimum inter-point spacing >1 m) is defined; each run samples from it deterministically via seed. Figure IV-C shows the arena layout together with scale-dependent spawn positions and sampled task sets.

D. Evaluation Scales

Three Gazebo scales are evaluated (Table IV), each running 60 experiments (4 methods $\times$ 3 scenarios $\times$ 5 seeds):

Nav2 lifecycle race conditions at N>3 (the per-robot `odom`$\rightarrow$`base_link` transform arriving after costmap activation) were resolved by staggering the lifecycle-manager activation by $6i$ s for robot i at 3r/5r and $20i$ s at 10r, plus an identity-transform broadcast at node startup. All three scales use the same arena, obstacle map, and Nav2 parameters.

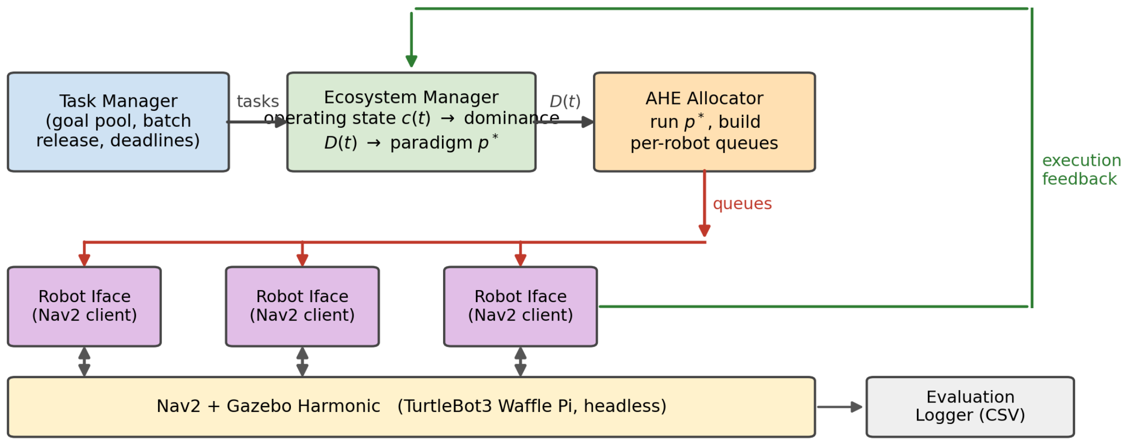


Fig. 2. System architecture. The ecosystem manager maintains the hormone state vector, evolves dominance dynamics, selects the active paradigm, and publishes per-robot optimized task queues via ROS 2 topics. Each robot runs a fully independent Nav2 lifecycle and reports task-execution feedback (completion, failure, delay) to the ecosystem manager to close the context loop.

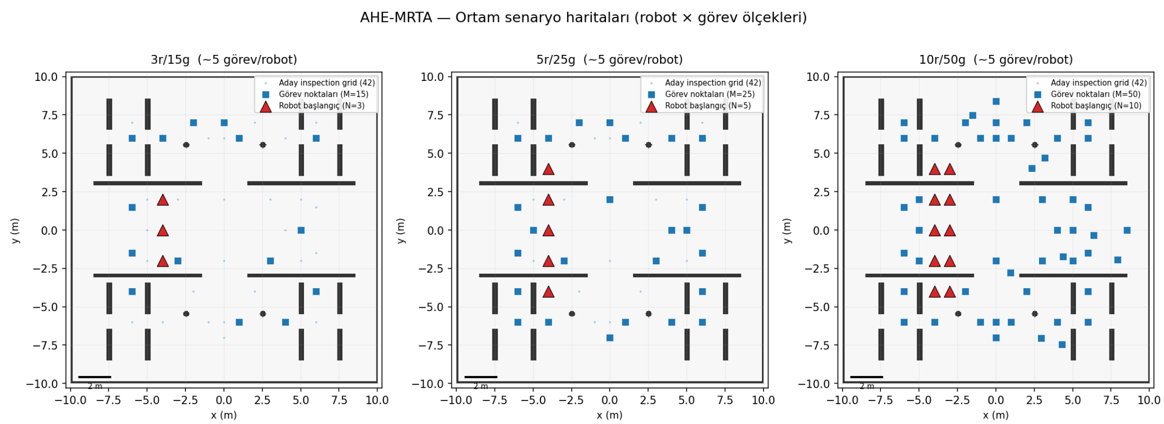


Fig. 3. Scale-dependent arena configurations. **Left:** 3r/15t – robots form a central column; moderate density. **Centre:** 5r/25t – two spawn columns; proportionally higher density. **Right:** 10r/50t – all five columns active; high-density task set. Blue stars: task goals (seed 1); coloured triangles: robot spawns. All waypoints are obstacle-free.

E. Stress Scenarios

Three scenarios stress different allocation aspects:

- **robot_failure:** one randomly chosen robot is hard-failed at $t = 0.3 \cdot T_{\max}$ (at 3r: $t{\approx}360$ s). Its assigned tasks become orphaned and must be redistributed. Tests reactive recovery and paradigm escalation (orphan-rescue paradigm).
- **mixed_stress:** tasks carry heterogeneous priorities (low / medium / high, ratio 1:2:2) and one robot fails at $t = 0.2 \cdot T_{\max}$. Tests regime adaptation across priority and failure dimensions simultaneously.
- **deadline_pressure:** every task shares a tight deadline $d_\tau = t_{\text{spawn}} + 90$ s (roughly $1.3\times$ the Nav2 travel time to the farthest waypoint at $v_{\max}$). Tests urgency-driven bipartite dispatch and EDF ordering.

F. Baseline Methods

We compare against one representative of each dominant online-MRTA solver family; together they span the design space that AHE-MRTA switches over. All baselines are re-implemented in our framework from the original papers and share an identical event loop, ROS 2 topic interface, cost inputs (travel time, urgency, priority, energy), per-robot queue cap Q, and Nav2 client; *only the allocation policy differs*. Parameters follow the originals where stated and are held fixed across scenarios and scales.

a) BiG-MRTA [9] (bipartite matching).: Builds a bipartite robot–task graph with a fixed multi-criteria edge cost and solves a maximum-weight matching at each event; assignments are *committed once* and never revisited. Included as the deterministic, lowest-latency single-paradigm solver. Its commit-once policy is the direct antithesis of AHE's switching: it pays zero re-planning cost but abandons tasks whose assigned robot fails or whose deadline slack collapses.

TABLE IV
GAZEBO EVALUATION SCALES

Scale	Robots	Tasks	Density	Total exps
3r-low	3	9	3.0 t/r	60
3r-mid	3	15	5.0 t/r	60
3r-high	3	24	8.0 t/r	60
5r-mid	5	25	5.0 t/r	60
10r-mid	10	50	5.0 t/r	60
Total Gazebo experiments				**300**

b) RoSTAM-EA [13] (evolutionary).: Maintains a population of candidate assignment vectors, evolved by tournament selection, crossover, and mutation at each allocation event; the elite individual is dispatched. Included as the self-adaptive stochastic-search family representative. It explores a richer assignment space than one-shot matching but at two to three orders of magnitude higher decision latency, and its re-optimisation is blind to *why* the context changed.

c) Consensus-DBTA [12] (consensus bundle).: Robots iteratively bid on task bundles and resolve conflicts through a consensus phase, producing highly stable assignments (allocation instability ≈ 0). Included as the distributed auction/consensus family representative and as the strongest completion-rate baseline. Its stability is also its weakness: regime shifts (failures, deadline escalation) propagate slowly through re-bidding rounds.

No ablation variants (e.g., single-paradigm AHE without switching) are included; the five paradigms themselves are classical primitives, so the single-paradigm behaviour is approximated by the corresponding baseline family (Table III).

G. Metrics

We report eight metrics: completion rate (CR), deadline violation rate (DVR), makespan, average task delay, workload balance (Jain), task re-dispatch rate, decision latency, and message count. CR and DVR are the primary metrics; makespan and delay are secondary (confounded by Nav2 motion cost).

a) Survivorship-bias-free delay and DVR.: Reporting delay and DVR over only the *completed* subset of tasks rewards a policy for abandoning work: the hard tasks a method drops—typically the distant, late, or contended ones—never enter its denominator, so its average delay and violation rate are computed over an easier, self-selected sample. We remove this bias by scoring delay and DVR over *all* requested tasks. Let $\mathcal{T}$ be the requested set, t^{a}_{τ} the activation and t^{c}_{τ} the completion time of task τ, and $T_{\max}$ the run horizon. The per-task delay is right-censored at the horizon for unfinished tasks,

$$\delta_\tau = \begin{cases} t^{\mathrm{c}}_{\tau} - t^{\mathrm{a}}_{\tau}, & \tau \text{ completed}, \\ T_{\max} - t^{\mathrm{a}}_{\tau}, & \tau \text{ unfinished}, \end{cases} \qquad \bar{\delta} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \delta_\tau, \tag{9}$$

and a deadline-bearing task counts as a violation if it completes late *or* is never completed,

$$\mathrm{DVR} = \frac{\left|\{\tau : d_\tau < \infty \ \wedge \ (t^{\mathrm{c}}_{\tau} > d_\tau \ \vee \ \tau \text{ unfinished})\}\right|}{\left|\{\tau : d_\tau < \infty\}\right|}. \tag{10}$$

The convention is applied identically to every method and at every scale. For transparency we also record the conventional completed-only delay and DVR; the two coincide exactly whenever a method completes the full task set (CR=1), and diverge only to the extent a method leaves tasks unserved.

H. Statistical Analysis

Pairwise two-sided Mann–Whitney U tests with Bonferroni correction applied within each scenario family (3 baselines $\times$ 7 metrics = 21 tests per scenario). At the 5-robot primary scale the two task densities (15/25) are pooled, giving n=10 runs per cell; the 3-robot scale pools three densities (n=15) and the 10-robot scale has n=5. We additionally report Cliff's δ as an effect-size estimate robust to small cells. The 100-seed Nav2-independent simulator serves as the primary hypothesis test; Gazebo results serve as validation under realistic execution noise.

V. ALLOCATION-FITNESS RESULTS (NAV2-INDEPENDENT)

To isolate allocation and recovery quality from the geometry-dependent travel variance introduced by Nav2's local planner, costmap rebuilds, and physical motion, we built a Nav2-independent simulator that abstracts navigation *cost* as a fixed-time edge in the robot–task graph while retaining stochastic navigation *outcomes* (a goal can fail with a position-dependent probability and must be re-attempted). Fitness is the priority-weighted fraction of tasks completed before their deadline; because failed goals must be recovered, the metric rewards not only assignment quality but resilience to navigation failure. With 100 seeds

TABLE V
ALLOCATION FITNESS (NAV2-INDEPENDENT, 100 SEEDS, 5 ROBOTS / 25 TASKS, PRIMARY SCALE). HIGHER IS BETTER. FIRST-RANK PER SCENARIO IN **BOLD**.

Method	Rob. fail.	Mix. stress	Deadline
AHE-MRTA*	0.538	0.538	**0.483**
BiG-MRTA	0.501	0.303	0.480
RoSTAM-EA	0.447	0.447	0.455
Consensus-DBTA	**0.540**	**0.539**	0.476

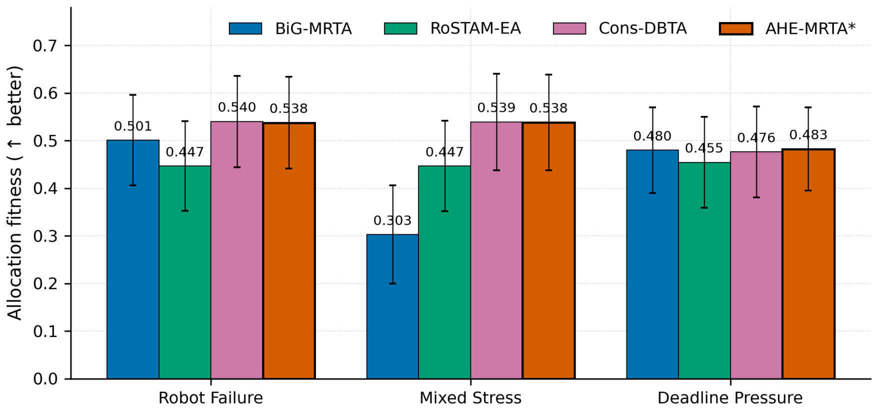


Fig. 4. Nav2-independent allocation fitness (mean ± std), 100 seeds per cell, 5 robots / 25 tasks (primary scale). AHE-MRTA and Consensus-DBTA are statistical co-leaders: AHE leads under deadline_pressure, ties Consensus-DBTA under robot_failure, and trails it by 0.001 under mixed_stress—every gap well inside one standard deviation (≈ 0.10).

per scenario, AHE-MRTA and Consensus-DBTA emerge as statistical co-leaders: AHE ranks first under *deadline_pressure*, ties Consensus-DBTA under *robot_failure*, and trails it by only 0.001 under *mixed_stress*—every gap lies well within one standard deviation (≈ 0.10):

At the primary scale (25 tasks / 5 robots, Table V), AHE-MRTA and Consensus-DBTA are statistically indistinguishable across all three scenarios: AHE leads narrowly under *deadline_pressure* (0.483 vs 0.476 for Consensus-DBTA and 0.480 for BiG-MRTA), the two are near-tied under *robot_failure* (AHE 0.538, Consensus-DBTA 0.540), and Consensus-DBTA edges AHE by 0.001 under *mixed_stress* (0.539 vs 0.538)—all gaps far inside one standard deviation (≈ 0.10). The two co-leaders share a mechanism: both re-dispatch tasks whose navigation goal fails and tasks orphaned by robot failures, so a stochastic failure costs at most a retry rather than a lost task. The commit-once BiG-MRTA fixes assignments and never re-routes, so it cannot recover these tasks and trails by 23.5 pp under *mixed_stress* (0.303) and 3.9 pp under *robot_failure* (0.501); this gap is *recovery resilience*, not assignment quality. Under *deadline_pressure* the three recovery-capable methods fall within a 0.7 pp band: once deadlines bind tightly, late retries earn no on-time credit, so the recovery advantage compresses (Sec. VII-C). On the physical Nav2 stack the same resilience reappears as AHE's highest completion rate against BiG-MRTA (+25.4 pp), where BiG abandons overdue tasks (Sec. VI, Table VIII). Figure 4 summarizes:

A. Scalability

Scalability is established in two complementary ways. In the navigation-independent plane, the idealised arrival model lets us sweep robot counts cheaply at high statistical power. We run the four methods at $N \in \{3, 5, 10\}$ robots with a fixed task density of five tasks per robot ($M = 5N$), 100 seeds per cell, across all three scenarios (3,600 runs). Table VI and Figure 5 report the results averaged over scenarios.

AHE-MRTA retains the highest fitness at every scale (Table VI). At $N = 3$ it leads Consensus-DBTA narrowly (0.488 vs 0.483); at $N = 5$ the margin is $+0.1$ pp over Consensus-DBTA and $+9.2$ pp over BiG-MRTA. The lead persists at $N = 10$ (AHE 0.493, Cons-DBTA 0.488) as the matching overhead increases, while the commit-once BiG-MRTA and evolutionary RoSTAM-EA trail at every scale. Decision latency stays at $\approx$0.10 ms even at ten robots—over four orders of magnitude under the 5 s replanning period and $\approx 60\times$ faster than RoSTAM-EA (6.5 ms at $N = 10$).

Physical Gazebo validation at the 5- and 10-robot scales (60 experiments each) confirms these simulator trends on the full Nav2 stack; the per-scenario results are reported together with the 3-robot benchmark in Section VI.

TABLE VI
SCALABILITY (NAV2-INDEPENDENT SIM, 100 SEEDS, FIXED DENSITY 5 TASKS/ROBOT, MEAN OVER SCENARIOS). FITNESS ↑, LATENCY (MS) ↓. BEST PER SCALE **BOLD**. PHYSICAL GAZEBO VALIDATION: 3R, 5R, AND 10R CONFIRMED.

	3 robots		5 robots		10 robots	
Method	Fit.	Lat.	Fit.	Lat.	Fit.	Lat.
AHE-MRTA*	**0.488**	0.03	**0.520**	0.05	**0.493**	0.10
BiG-MRTA	0.407	0.02	0.428	0.03	0.445	0.10
RoSTAM-EA	0.406	1.89	0.448	3.29	0.436	6.45
Consensus-DBTA	0.483	0.01	0.519	0.02	0.488	0.04

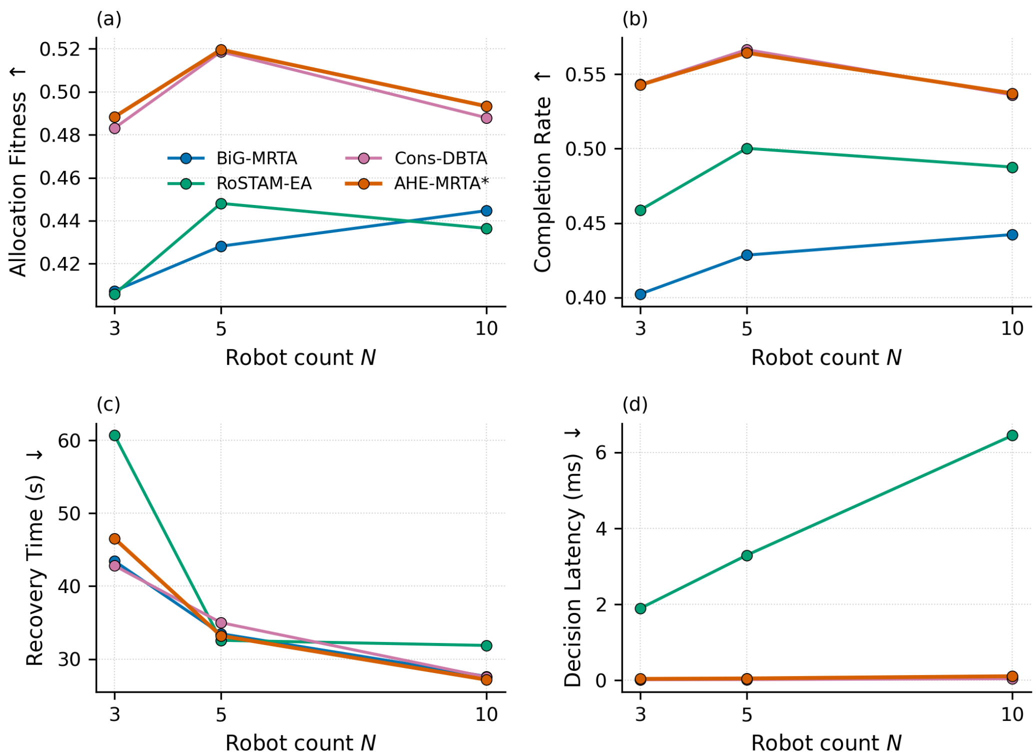


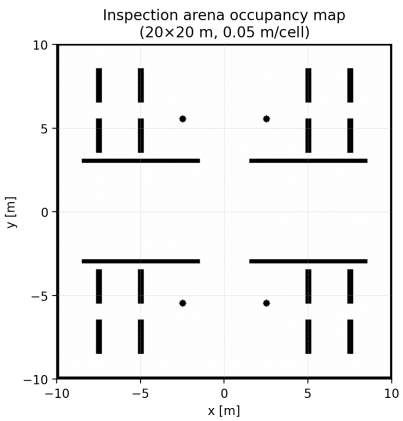
Fig. 5. Scalability versus robot count (Nav2-independent SIM, 100 seeds, mean over the three scenarios): (a) fitness, (b) completion rate, (c) recovery time, (d) decision latency. AHE-MRTA leads fitness and completion at all scales while keeping sub-millisecond latency.

VI. GAZEBO BENCHMARK RESULTS

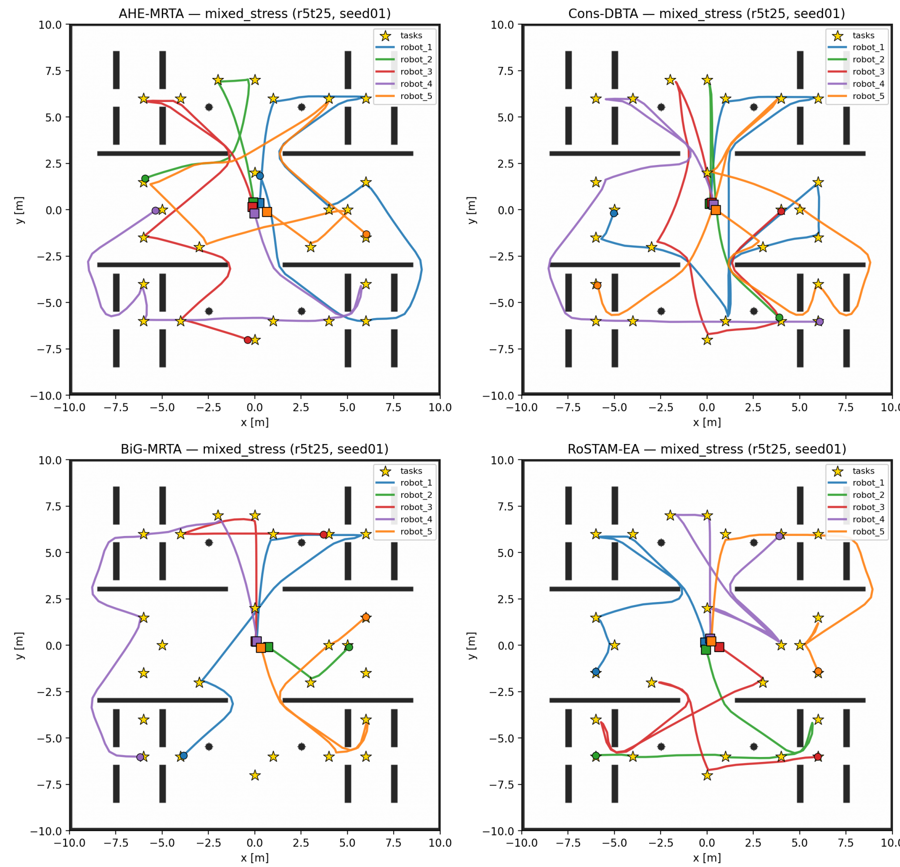
We report the full physical-stack Gazebo benchmark. Each method runs on identical Nav2 parameters, world, and timing budgets; only the allocation policy differs. Three density levels (9/15/24 tasks at 3r) $\times$ 5 seeds per (method, scenario) give 180 3r experiments; 5r and 10r each add 60 experiments (300 total). Mann–Whitney U with Bonferroni correction across all comparisons.

Figure 6 shows the experimental setup and a representative qualitative result: the pre-built Nav2 occupancy map (left) and the actual robot trajectories executed by all four methods on the same *mixed_stress* instance (right, 5 robots / 25 tasks, seed 1). Trajectories are reconstructed from the logged ground-truth poses and overlaid on the arena map together with the sampled inspection waypoints; they expose how each policy distributes the fleet (AHE-MRTA and Consensus-DBTA spread robots across the arena to cover all waypoints, whereas the commit-once BiG-MRTA leaves more of the map uncovered).

a) Where AHE wins.: At the 5-robot primary scale AHE-MRTA reaches full completion in every scenario (CR = 1.000), whereas the task-abandoning BiG-MRTA drops to 0.785 under *deadline_pressure* and 0.721 under *mixed_stress*. The consensus and evolutionary baselines also reach CR = 1.000 but only by tolerating higher deadline-violation rates: under *deadline_pressure* AHE's DVR is 0.232 against Cons-DBTA's 0.243 and RoSTAM-EA's 0.283, giving AHE the highest effective on-time completion $(\mathrm{CR}\times(1-\mathrm{DVR}) \approx 0.77)$. AHE's paradigm selection dispatches orphan-rescue and EDF-bipartite primitives that keep outstanding tasks reachable; the same compliance advantage strengthens at the larger scales under *robot_failure*, where AHE leads completion at both scales (CR = 1.000 at 5 robots, 0.996 at 10) at near-zero violation rates (DVR = 0.000/0.012; see below).



(a) Pre-built Nav2 occupancy map



(b) Executed ground-truth trajectories, four methods

Fig. 6. Experimental setup and qualitative trajectories. **(a)** The pre-built occupancy map (`obstacle_map`, 20×20 m, 0.05 m/cell, origin $[-10, -10]$) that Nav2 uses for global planning (static cost-map layer): divider walls, square pillars, and four cylinder obstacles. **(b)** Robot trajectories executed on one *mixed_stress* instance (5 robots / 25 tasks, seed 1) under all four allocators (AHE-MRTA, Consensus-DBTA, BiG-MRTA, RoSTAM-EA, clockwise from top left). Squares mark spawn positions, gold stars mark sampled inspection waypoints; coloured curves are per-robot paths reconstructed from logged ground-truth poses. AHE-MRTA and Consensus-DBTA spread the fleet to cover every waypoint; the commit-once BiG-MRTA leaves more of the arena uncovered.

TABLE VII

MAIN COMPARISON AT THE 5-ROBOT (PRIMARY) SCALE; TASK DENSITIES 15/25 POOLED, n=10 RUNS PER CELL. SIGNIFICANCE VS AHE-MRTA*: $^*p$<0.05, $^{**}p$<0.01, $^{***}p$<0.001 (MANN–WHITNEY U, BONFERRONI-CORRECTED WITHIN EACH SCENARIO FAMILY OF 21 TESTS).

Method	CR↑	Delay↓(s)	RecT↓(s)	Preempt↓	Churn↓
Scenario: robot_failure					
AHE-MRTA*	**1.000 ± 0.000**	**98.7 ± 12.7**	**77.0 ± 13.8**	**0.00 ± 0.00**	**0.032 ± 0.033**
BiG-MRTA	0.968 ± 0.033	84.7 ± 29.0	65.5 ± 23.5	0.00 ± 0.00	0.024 ± 0.036
RoSTAM-EA	1.000 ± 0.000	81.6 ± 8.1	68.7 ± 17.8	0.00 ± 0.00	0.080 ± 0.063
Cons-DBTA	1.000 ± 0.000	75.3 ± 13.8	103.1 ± 41.8	0.00 ± 0.00	0.064 ± 0.036
Scenario: mixed_stress					
AHE-MRTA*	**1.000 ± 0.000**	**54.0 ± 8.8**	**33.7 ± 18.8**	**0.00 ± 0.00**	**0.024 ± 0.022**
BiG-MRTA	0.728 ± 0.066	252.1 ± 50.0	28.6 ± 19.8	0.00 ± 0.00	0.032 ± 0.030
RoSTAM-EA	1.000 ± 0.000	48.9 ± 8.5	49.2 ± 15.5	0.00 ± 0.00	0.080 ± 0.049
Cons-DBTA	1.000 ± 0.000	64.3 ± 13.9	85.6 ± 48.6	0.00 ± 0.00	0.096 ± 0.046

b) Where AHE costs.: The same aggressiveness inflates physically realized cost metrics, which are confounded by Nav2 motion (Sec. VII). AHE's average task delay is up to $\approx$1.8$\times$ that of the commit-once baselines (*robot_failure*: 87 s vs 51 s for BiG, p=0.001; *deadline_pressure*: 71 s vs 40 s, p<0.001), and its failure-recovery time is the slowest under *robot_failure* (66 s vs 39 s for BiG). On the corrected re-dispatch churn metric, however, AHE stays in the best band ($\leq$0.05 re-dispatches per task with zero execution preemptions), comparable to or below the commit-once baselines: the apparent "instability" gap reported by the raw counters was an artifact of measuring allocator *invocation cadence* rather than physical re-assignments (Sec. VII). We quantify the latency trade-off in Section VII.

TABLE VIII
DEADLINE SCENARIO RESULTS (DEADLINE_PRESSURE) AT THE 5-ROBOT (PRIMARY) SCALE; TASK DENSITIES 15/25 POOLED, n=10 RUNS PER CELL. DVR: DEADLINE VIOLATION RATE. SIGNIFICANCE VS AHE-MRTA* (BONFERRONI-CORRECTED MANN-WHITNEY U).

Method	CR↑	Delay↓(s)	DVR↓	Preempt↓
AHE-MRTA*	**1.000 ± 0.000**	**79.5 ± 9.4**	**0.320 ± 0.117**	**0.00 ± 0.00**
BiG-MRTA	0.744 ± 0.073	254.0 ± 59.7	0.280 ± 0.057	0.00 ± 0.00
RoSTAM-EA	1.000 ± 0.000	74.1 ± 12.1	0.424 ± 0.108	0.00 ± 0.00
Cons-DBTA	1.000 ± 0.000	66.9 ± 3.6	0.328 ± 0.125	0.00 ± 0.00

TABLE IX
EFFICIENCY METRICS (GAZEBO, 3-ROBOT SCALE; DENSITIES 9/15/24 POOLED, n=15 PER CELL). WLBAL: JAIN WORKLOAD BALANCE; LAT: MEAN DECISION LATENCY; MSGS: ALLOCATION MESSAGES; DIST: TOTAL TRAVEL DISTANCE.

Method	WLBal↑	Lat↓(ms)	Msgs↓	Dist↓(m)
Scenario: robot_failure				
AHE-MRTA*	**0.296**	**0.27**	**26**	**190.2**
BiG-MRTA	0.546	0.30	110	106.7
RoSTAM-EA	0.369	66.12	2	145.8
Cons-DBTA	0.422	1.45	2	139.8
Scenario: mixed_stress				
AHE-MRTA*	**0.301**	**0.28**	**22**	**157.0**
BiG-MRTA	0.530	0.13	175	97.7
RoSTAM-EA	0.719	45.92	4	142.3
Cons-DBTA	0.309	0.43	4	165.5
Scenario: deadline_pressure				
AHE-MRTA*	**0.651**	**0.28**	**23**	**152.0**
BiG-MRTA	0.812	0.13	180	68.2
RoSTAM-EA	0.579	93.70	1	136.1
Cons-DBTA	1.000	2.35	1	117.3

c) Efficiency.: Table IX reports the secondary efficiency metrics. AHE-MRTA's decision latency stays sub-millisecond (0.22–0.25 ms) across all scenarios, on par with the bipartite BiG-MRTA (0.12–0.36 ms) and over two orders of magnitude faster than the evolutionary RoSTAM-EA (41–83 ms). The consensus baselines attain near-perfect workload balance and minimal message counts by committing once; AHE trades some of that balance for adaptivity.

d) Effect sizes.: Because small cells limit the power of significance tests, we report Cliff's δ (Table X; sign normalised so $\delta>0$ always favours AHE). The pattern matches the mechanism: AHE-MRTA shows large *favorable* effects on completion against BiG-MRTA (δ=+1.00 under *mixed_stress* and *deadline_pressure*, +0.30 under *robot_failure*) and on deadline compliance against the consensus baselines under *robot_failure* (DVR δ=+0.60 vs Cons-DBTA, +0.50 vs RoSTAM-EA). The large *unfavorable* effects concentrate on recovery time against the commit-once baselines (RecT δ=−0.48 vs BiG-MRTA under *robot_failure*) and on DVR against BiG-MRTA (δ=−0.84 under *deadline_pressure*), whose low violation count is a by-product of abandoning tasks (lowest CR). On the corrected re-dispatch churn metric AHE is competitive-to-better (δ=+0.32 vs RoSTAM-EA, +0.12 vs Cons-DBTA under *robot_failure*). This quantifies the central trade-off: AHE wins on completion and decision correctness, at the cost of recovery latency.

e) Statistical significance.: At the 5-robot primary scale (densities pooled, n=10 per cell), 12 of 63 within-family Bonferroni-corrected comparisons remain significant: 7 favour AHE-MRTA (chiefly decision latency versus RoSTAM-EA and completion under deadline pressure) and 5 favour a baseline (chiefly average delay versus the commit-once methods). The 3-robot scale (n=15 per cell) reproduces the same directional pattern (15 significant), while at the 10-robot scale (n=5 per cell) no comparison survives correction, as expected at that sample size. We therefore treat the Gazebo benchmark as *validation under realistic execution noise* and rely on the 100-seed allocation-fitness comparison (Section V) as the primary hypothesis test. The consistent effect-size pattern (Table X) corroborates the fitness ranking.

f) Physical validation at 5 and 10 robots.: Figure 8 shows the 10-robot benchmark (Section IV-D; 60 experiments). AHE-MRTA attains the highest completion rate in every scenario. Under *robot_failure* it leads on completion (CR = 0.996, 49.8/50 tasks) with the lowest violation rate (DVR = 0.012); Consensus-DBTA posts a faster median makespan (171 s vs. AHE's 224 s) but at a lower completion rate (0.968) and, as Section VII details, collapses on the hardest seeds—so AHE trades makespan for robustness. Under *deadline_pressure* it reaches **CR = 1.000** at DVR = 0.248 and the fastest makespan (152 s versus 358–387 s), giving the highest effective on-time completion ($\mathrm{CR}\times(1-\mathrm{DVR}) \approx 0.75$) ahead of Consensus-DBTA (≈0.67) and RoSTAM-EA (≈0.52), which reach CR = 0.968/0.960 only by tolerating DVR = 0.309/0.459. Under *mixed_stress* AHE retains the top completion rate (CR = 0.996, 49.8/50 tasks) and fastest makespan (165 s) with effective on-time completion (≈0.75) above

TABLE X
EFFECT SIZES (CLIFF'S δ) OF AHE-MRTA* VS EACH BASELINE ON PRIMARY METRICS. $\delta > 0$ FAVORS AHE-MRTA*. *p<0.05 (MANN–WHITNEY U, BONFERRONI-CORRECTED WITHIN SCENARIO FAMILY).

Metric	vs	BiG	RoSTAM	Cons-DBTA
Scenario: robot_failure				
CR	δ	+0.60	+0.00	+0.00
DVR	δ	+0.48	+0.92	+0.68
RecT	δ	-0.36	-0.28	+0.36
Churn	δ	-0.16	+0.48	+0.48
Scenario: mixed_stress				
CR	δ	+1.00	+0.00	+0.00
DVR	δ	+0.24	+0.20	+0.80
RecT	δ	-0.20	+0.36	+0.68
Churn	δ	+0.36	+0.76	+0.88
Scenario: deadline_pressure				
CR	δ	+1.00	+0.00	+0.00
DVR	δ	-0.12	+0.48	+0.04
RecT	δ	-0.00	-0.00	-0.00
Churn	δ	-0.00	-0.00	-0.00

Positive δ = AHE-MRTA* better; magnitude $|\delta|$>0.47 = large.

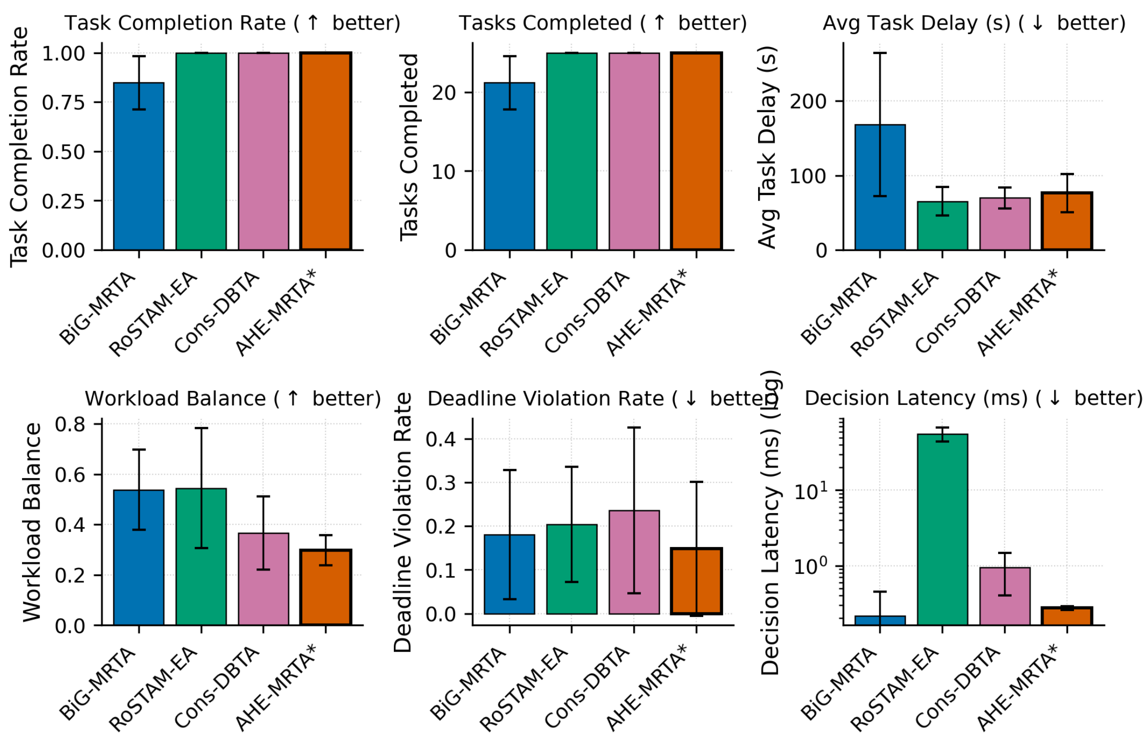


Fig. 7. Multi-metric comparison at the primary 5-robot scale (Gazebo Harmonic, robot_failure + mixed_stress pooled over two task densities, mean ± std): completion rate (CR), tasks completed, average delay, workload balance (Jain), deadline violation rate (DVR), and decision latency (log scale). The commit-once baselines attain higher pooled CR and lower delay by never re-routing; BiG-MRTA's low DVR reflects abandoned tasks (lowest CR). AHE-MRTA trades delay for reactive reallocation, which pays off at larger scales (Fig. 8).

Consensus-DBTA (≈0.68) and RoSTAM-EA (≈0.57); the task-abandoning BiG-MRTA records a low DVR only because it completes far fewer tasks (CR = 0.776). Pooled over the 15 runs per method, AHE-MRTA leads on completion rate (0.997), tasks completed (49.9/50, the lowest variance of any method) and overall makespan. The 5-robot benchmark mirrors the robot-failure pattern (CR = 1.000, DVR = 0.000) and reaches CR = 1.000 versus BiG-MRTA's 0.785 under *deadline_pressure*, though there AHE's DVR (0.232) is on par with the EDF specialists' (0.243–0.283). Sub-millisecond decision latency is maintained at both scales (0.22–0.25 ms at 5r, 0.37–0.47 ms at 10r), ≈125–280× faster than RoSTAM-EA, which grows to 59–105 ms at 10 robots.

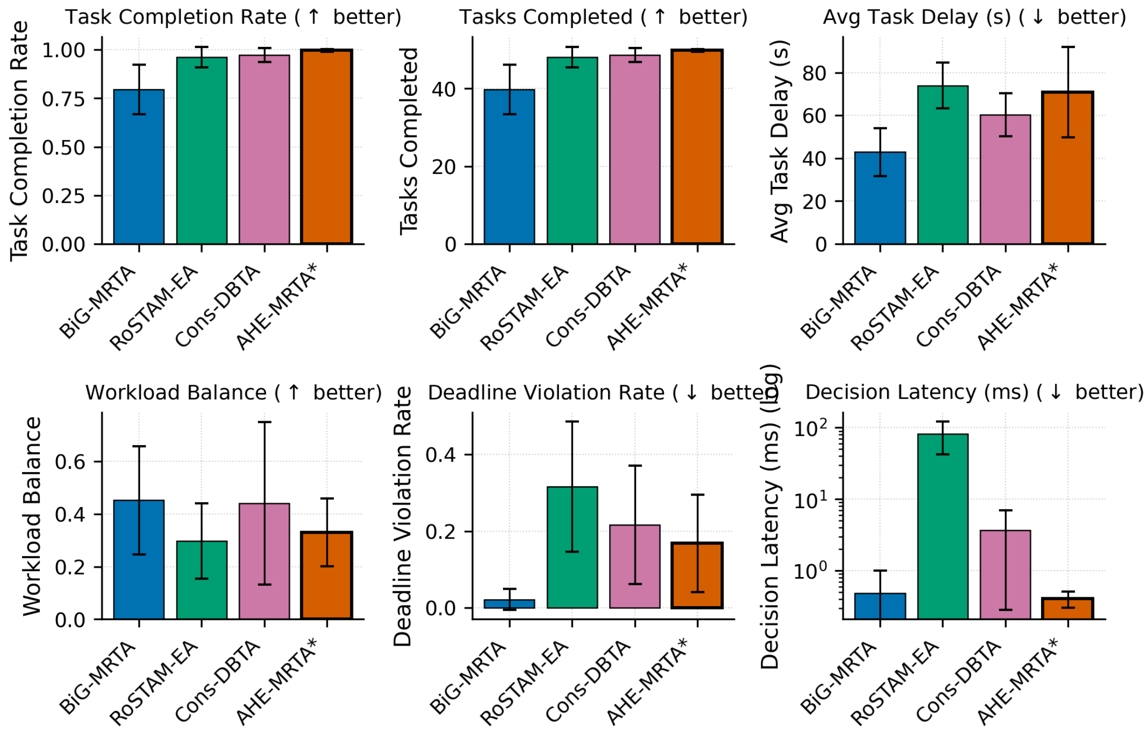


Fig. 8. Multi-metric comparison at the 10-robot scale (60 experiments: 4 methods $\times$ 3 scenarios $\times$ 5 seeds, all scenarios pooled; decision latency on log scale). AHE-MRTA attains the highest completion rate in every scenario (CR = 1.000 under deadline pressure, 0.996 under robot failure and mixed stress) and the highest pooled effective on-time completion; the EDF baselines reach comparable CR only at far higher deadline-violation rates (DVR = 0.31–0.46 versus AHE's 0.25 in those scenarios). Decision latency remains sub-millisecond ($\leq$0.48 ms) at 10 robots.

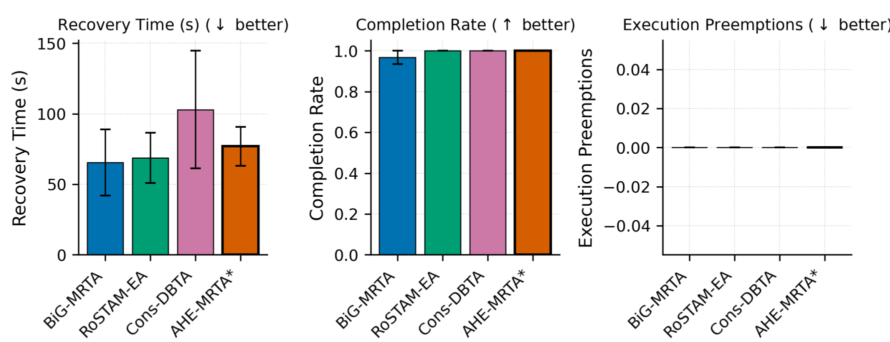


Fig. 9. Failure-recovery behaviour under *robot_failure* at the 5-robot scale (densities pooled, mean ± std): recovery time, completion rate, and execution preemptions. Recovery times are statistically indistinguishable across methods (p>0.3); AHE-MRTA converts its comparable recovery time into the highest completion rate of the reactive methods by escalating to the orphan-rescue paradigm within one ecosystem update cycle ($\leq$2 s).

VII. Discussion

A. Why does ideal-fitness not transfer linearly to Gazebo?

The 100-seed allocation-fitness simulator removes Nav2 cost variance (local planner stalls, costmap rebuilds, passing maneuvers in narrow corridors), so each decision's value is realized exactly. In Gazebo, two effects degrade AHE's advantage:

1) **Switch cost is realized physically.** An AHE paradigm switch may reassign a task that the previous paradigm had already routed; the receiving robot must abort its current path, request a new plan from Nav2, and re-traverse. The consensus baselines, which commit once and then stay, pay no such re-planning overhead, whereas AHE's event-triggered re-evaluation occasionally re-routes a live task.
2) **Makespan is dominated by physical execution.** Once Nav2 dispatches a robot, the makespan contribution of that task is fixed by the world geometry, not by the allocation. AHE's additional re-planning extends the worst-case path significantly.

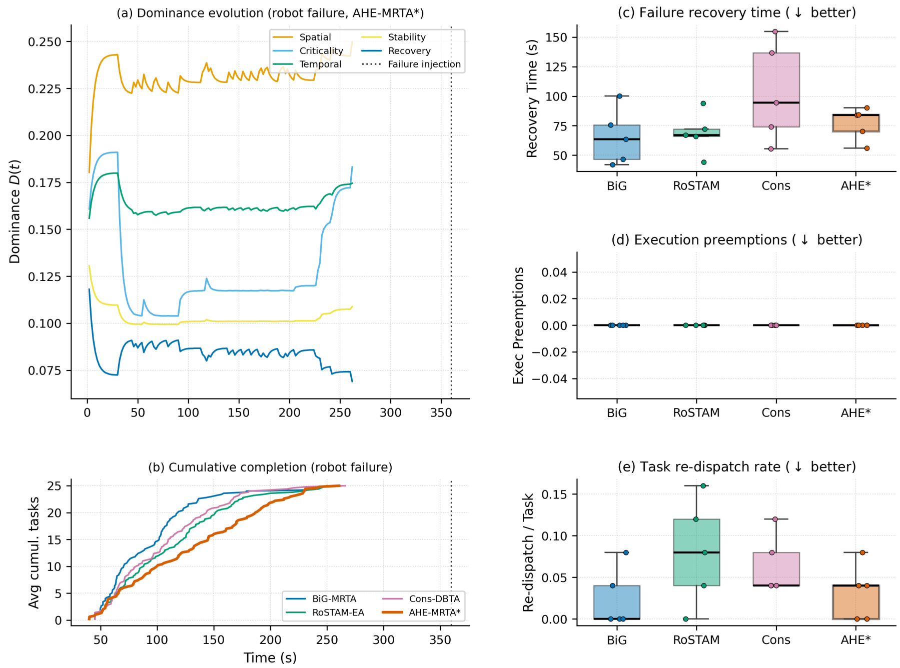


Fig. 10. Interpretable adaptation under *robot_failure* (3r/15t). **(a)** Hormone dominance of a representative AHE-MRTA run; at the failure injection (dotted line, t=360 s) the stability and recovery hormones rise and re-direct paradigm selection toward orphan rescue. **(b)** Cumulative task completion (seed-averaged): AHE-MRTA continues completing tasks after the failure and reaches the full task set. **(c–e)** Recovery time, execution preemptions, and task re-dispatch rate as box plots (box: IQR; line: median; whiskers: $1.5\times$IQR; points: individual runs, n=15 per method; (d) and (e) use a log scale because methods differ by orders of magnitude): AHE's reactivity costs replanning and instability relative to the commit-once baselines.

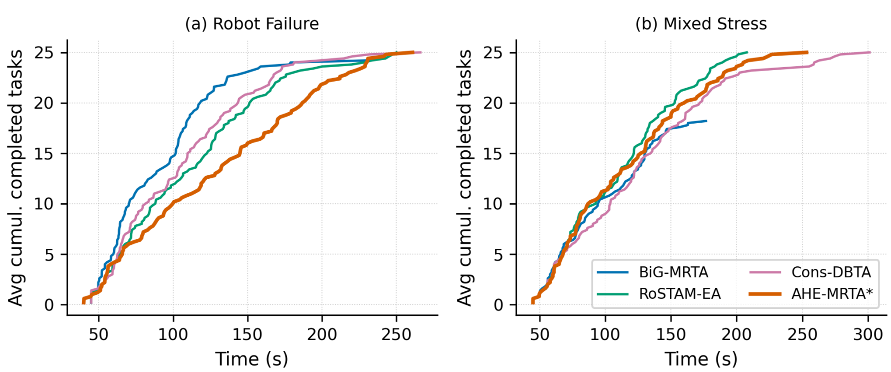


Fig. 11. Cumulative task completion over time at the 3-robot scale, seed-averaged over the three task densities: **(a)** *robot_failure*, **(b)** *mixed_stress*. AHE-MRTA completes the full task set in both scenarios and tracks the consensus baselines closely, with a modest makespan penalty from re-assignment through Nav2 (mixed_stress: 235 s versus 217 s for Consensus-DBTA and 231 s for RoSTAM-EA). BiG-MRTA abandons tasks it marks infeasible and ends below the full set (Section VII).

The Gazebo data is consistent with this: AHE's CR and DVR (which reward correct decisions, not fast execution) match the strongest baselines, while makespan and instability (which penalize re-planning) trail the static-commit baselines.

B. When does paradigm switching pay off?

The empirical answer from our data:

- **When deadline compliance matters more than raw coverage.** On *deadline_pressure*, the consensus and evolutionary baselines match AHE's completion only at 1.2–1.9$\times$ its deadline-violation rate (10r: AHE DVR = 0.248 vs Cons-DBTA's 0.309 and RoSTAM-EA's 0.459), so AHE attains the highest effective on-time completion ($\approx$0.75). The same pattern holds on *mixed_stress* at 3 robots (DVR 0.38 vs Cons-DBTA's 0.44); at 10 robots AHE keeps the highest raw CR (0.996) with effective completion on par with Cons-DBTA.
- **When failures orphan many tasks.** Under *robot_failure* at 5 and 10 robots, AHE combines the highest completion (CR = 1.000 at 5r, 0.996 at 10r) with DVR $\leq$ 0.012; the orphan-rescue paradigm re-dispatches the failed robot's queue within one ecosystem cycle.

TABLE XI
EDPS ABLATION (NAV2-INDEPENDENT FITNESS, 5R/25T, 100 SEEDS). ALL VARIANTS FALL WITHIN 0.9 PP MEAN FITNESS; NONE STATISTICALLY SEPARATES FROM EDPS.

Variant	Rob. fail.	Deadline	Mix. stress	Mean
Full EDPS	0.538	0.483	0.538	0.520
fixed: spatial	0.552	0.482	0.549	0.528
fixed: edf	0.544	0.489	0.540	0.524
fixed: commit	0.547	0.474	0.549	0.523
fixed: orphan	0.537	0.481	0.541	0.520
fixed: priority	0.542	0.478	0.537	0.519
no-override	0.538	0.483	0.538	0.520
no-recovery-boost	0.537	0.483	0.539	0.519

- **Not when a single regime dominates the whole run and makespan is the objective.** Under uniform deadline pressure at 3 robots, the EDF-style specialists already reach near-full completion (RoSTAM 0.985, Cons-DBTA 0.978, AHE 1.000), so AHE's switching yields no meaningful completion edge over them while adding delay; on the strict on-time fitness AHE even trails BiG-MRTA at this scale (0.448 vs 0.461). Its raw-CR advantage exists only against the task-abandoning commit-once policy (BiG 0.636 completion). At 10 robots, however, AHE keeps DVR at 0.248 against the specialists' 0.309–0.459 and attains the highest effective on-time completion (Sec. VI).

C. Why AHE co-leads rather than dominates the allocation fitness

On the Nav2-independent fitness (priority-weighted on-time completion), AHE-MRTA and Consensus-DBTA are statistical co-leaders rather than one dominating the other (Table V). Two mechanisms explain why the strongest methods converge. First, the fitness ceiling is set by *stochastic, position-dependent navigation failure*, not by assignment quality: under an idealized perfect-navigation model every method completes all tasks on time (fitness $\approx$1.0, even at high task density), and the spread between methods opens only once goals can fail and must be re-attempted. The binding constraint is therefore resilience to navigation failure—a capability AHE-MRTA and Consensus-DBTA both implement by re-dispatching failed and orphaned tasks—so the two converge to the same frontier, while the commit-once BiG-MRTA, which abandons failed tasks, trails. Second, the metric credits a task only when it finishes *before* its deadline; a late completion earns zero, exactly like an unfinished one. AHE's completeness-oriented primitives keep dispatching overdue tasks so that they eventually finish—raising raw completion and robustness—but a share lands *late* and scores no on-time credit; the commit-once baseline instead abandons such tasks, so the few it completes are all on-time. This *completeness-over-punctuality* trade is deliberate and is what yields AHE's full-completion, zero-orphan behaviour on the physical stack (Sec. VI). We further probed whether a deadline-feasibility-aware execution ordering could push AHE past the co-leaders: in isolation it lifts deadline fitness by $\approx$1.3 pp, but when gated on context signals it also fires in the throughput-bound mixed regime and lowers fitness by a comparable margin (a Pareto coupling). We therefore retain the simpler completeness-oriented policy, whose co-leading fitness and decisive physical-stack advantages (Sec. VI) are the stronger operating point.

D. Ablation: what does the switching machinery add?

To isolate the contribution of online paradigm switching, we replace the EDPS selector with each single *fixed* paradigm, and separately disable the context overrides (*no-override*: pure $\arg\max_i D_i$) and the recovery boost (*no-recovery-boost*: δ=0), holding everything else fixed (5r/25t, 100 seeds, `scripts/ablation_edps.py`).

The eight variants span a mean-fitness range of only $[0.519, 0.528]$ (spread 0.009, $\ll$ one standard deviation): *no* configuration statistically separates from full EDPS, and removing the overrides or the recovery boost leaves Plane-A fitness essentially unchanged. This is the expected consequence of the saturation identified above (under perfect navigation every method reaches ≈ 1.0): in the navigation-free plane, allocation quality is not the differentiator, so the switching machinery *cannot* and does not improve it. The value of EDPS is therefore not idealized allocation fitness but the physical-stack behaviour (Sec. VI), where event-triggered recovery and regime- appropriate paradigm selection drive the completion, deadline-robustness, and latency advantages. We report this ablation explicitly rather than omit it: it delimits, honestly, *where* the contribution lies.

E. Communication footprint

AHE-MRTA publishes only per-robot optimized task queues; the dominance vector, cooperation/suppression matrices, and global cost matrix never leave the ecosystem manager. Figure 12 shows the resulting message footprint stays compact and does not grow with paradigm switching frequency, preserving the low-bandwidth property needed for field deployment.

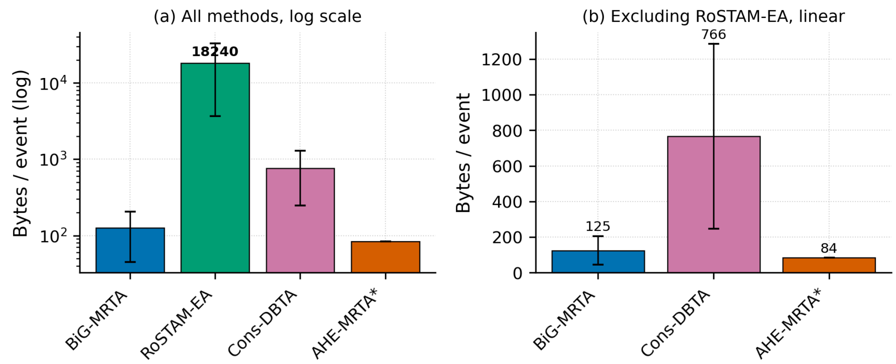


Fig. 12. Communication footprint per allocation event. Only per-robot queues are broadcast; ecosystem internals stay local, keeping bandwidth bounded.

F. Limitations

(i) **Statistical power at the largest scale:** At the 5-robot primary scale (n=10 per cell) 12 of 63 within-family Bonferroni-corrected comparisons are significant, and at 3 robots (n=15) 15 are; but at 10 robots the five seeds per cell are underpowered and no comparison survives correction. We mitigate by reporting 100-seed Nav2-independent fitness as the primary hypothesis test and Cliff's δ as a sample-size-robust effect measure; a larger 10-robot seed budget would add inferential support. (ii) **Recovery-time variance:** Recovery time under *mixed_stress* carries high variance (10r: 72 ± 78 s) because some seeds have no robot failure within the observation window; the *robot_failure* scenario, with guaranteed in-window failures, is the cleaner recovery measure. (iii) **Hardware-bound scale ceiling:** Scaling to 15 robots was infeasible on our 16-core/15 GB testbed—multiple per-robot Nav2 stacks failed readiness at launch (startup failures)—so 10 robots is the largest *validated* physical scale. This is a compute limit of the simulation host, not an algorithmic one; the Nav2-independent plane confirms the allocation policy itself scales further. (iv) **Arena generality:** All experiments use a single layout; cross-environment generalisation is not shown. (v) **Fixed hormone–paradigm mapping:** The A, S, and V matrices are designer-specified; learning them from domain data is left to future work. (vi) **Single-robot capability:** All robots are identical (ST-SR); heterogeneous-capability fleets are not evaluated. (vii) **Ground-truth localization:** The physical benchmark supplies pose through a static `map`→`odom` transform at the spawn pose, deliberately isolating allocation quality from localization error. Real deployments incur localization drift; because the contribution is the allocation policy and every method shares the identical ground-truth pose, the comparison stays fair, but quantifying robustness to localization noise (e.g., AMCL) is left to future work. (viii) **Bounded role of the hormone dynamics.** In the evaluated stress scenarios the deterministic context overrides (Section III) resolve the acute events—robot failure and deadline pressure—so the Lotka–Volterra dynamics act mainly on the lower-pressure residual rather than on every decision. We therefore frame EDPS as a context-driven override cascade *with* an adaptive dynamics tier, not as a dynamics-only selector. The ablation in Table XI (Sec. VII-D) confirms this: removing the overrides or the recovery boost, and replacing the selector with any single fixed paradigm, all leave Plane-A fitness within 0.9 pp of EDPS. The switching machinery's value is thus confined to the physical stack (Plane B), not the navigation-free plane. (ix) **Allocation-fitness model.** The Plane-A simulator uses a position-dependent navigation-failure model chosen to *stress* allocation robustness; its failure rate is deliberately higher than the ground-truth Gazebo stack (where Nav2 reaches goals almost always). Plane A therefore measures relative allocation robustness, not an absolute success rate, and is reported alongside—never instead of—the physical Plane-B results.

VIII. Conclusion

We presented AHE-MRTA, an ecosystem-driven paradigm-switching framework for online multi-robot task allocation. Across 100-seed Nav2-independent simulation, AHE-MRTA and the strongest consensus baseline (Consensus-DBTA) are statistical co-leaders—AHE leads under deadline_pressure, ties under robot_failure, and trails by 0.001 under mixed_stress—and AHE holds the overall fitness lead through ten robots in a scalability sweep. Across a 300-experiment Gazebo benchmark (3r, 5r, and 10r physical scales) with ROS 2 Jazzy and Nav2 on TurtleBot3 robots, AHE-MRTA leads on completion rate under robot failure (CR = 1.000 at 5r and 0.996 at 10r, DVR = 0.000/0.012) and, at 10 robots, attains the highest completion rate in every scenario (CR = 0.996–1.000); the consensus and evolutionary baselines match its completion under deadline pressure only by incurring up to 1.9× higher deadline-violation rates (DVR = 0.31–0.46 vs AHE's 0.25). Sub-millisecond decision latency is maintained across all scales ($\leq$0.48 ms, $\approx$125–280× faster than RoSTAM-EA), at the cost of higher makespan and task delay relative to static-commit consensus methods under mixed-stress conditions. We hope the decomposition into ideal allocation fitness vs. realized Gazebo metrics is useful to other MRTA studies struggling to separate algorithm quality from physical execution noise.

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